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Home > IP Routing Quiz

IP Routing Quiz

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Result of IP Routing Quiz:

Total Questions	Full Score	Passing Rate	Your Score	Correct Answer Percentage	Elapsed
14	150	80%	150	100%	00:01:46

Congratulations!

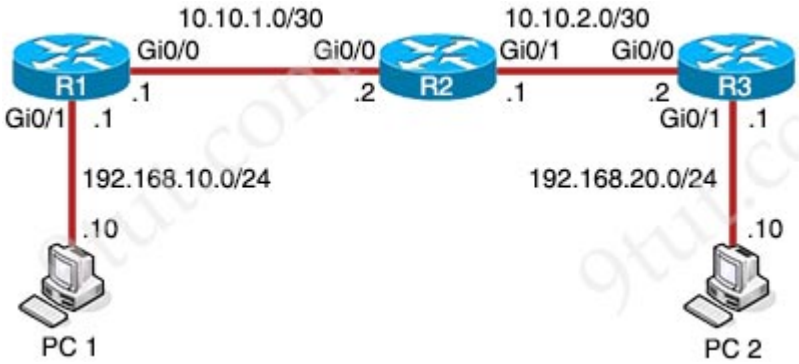
You passed this test!

If you want to retake this quiz, please press Ctrl + F5 on Windows or press CMD + R on Mac.

Your answers are shown below:

Question 1

Refer to the exhibit. When PC 1 sends a packet to PC2, the packet has which source and destination IP address when it arrives at interface Gi0/0 on router R2?



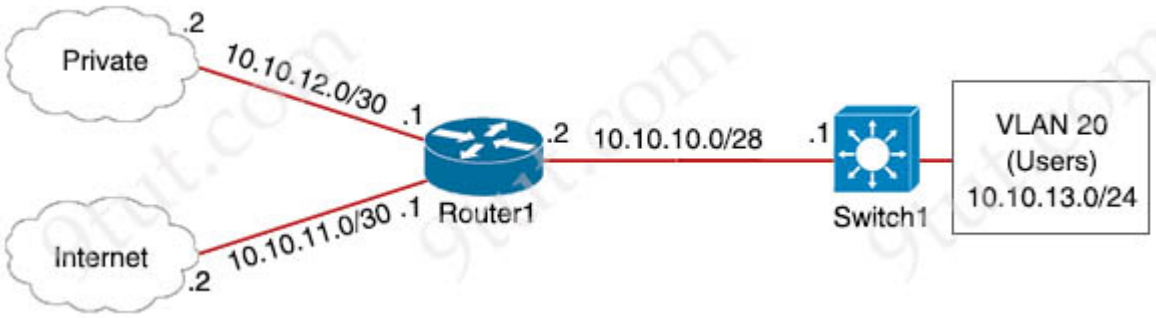
- A. source 192.168.20.10 and destination 192.168.20.1
- **B. source 192.168.10.10 and destination 192.168.20.10**
- C. source 10.10.1.1 and destination 10.10.2.2
- D. source 192.168.10.10 and destination 10.10.2.2

Explanation

The source and destination IP addresses of the packets are unchanged on all the way. Only source and destination MAC addresses are changed.

Question 2

Refer to the exhibit. Which path is used by the router for Internet traffic?




```
R1#show ip route
Gateway of last resort is 10.10.11.2 to network 0.0.0.0
209.165.200.0/27 is subnetted, 1 subnets
B       209.165.200.224 [20/0] via 10.10.12.2, 00:10:34
        10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C               10.10.10.0/28 is directly connected, GigabitEthernet0/0
C               10.10.11.0/30 is directly connected, FastEthernet2/0
C               10.10.13.0/30 [110/2] via 10.10.10.1, 00:03:34, GigabitEthernet0/0
C               10.10.12.0/30 is directly connected, GigabitEthernet0/1
S*      0.0.0.0/0 [1/0] via 10.10.11.2

Switch1#show ip route
Gateway of last resort is not set
        10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C               10.10.10.0/28 is directly connected, FastEthernet0/1
C               10.10.13.0/24 is directly connected, VLAN20
```

- A. 10.10.13.0/24
- **B. 0.0.0.0/0**
- C. 209.165.200.0/27
- D. 10.10.10.0/28

Question 3

When a floating static route is configured, which action ensures that the backup route is used when the primary route falls?

- A. The floating static route must have a lower administrative distance than the primary route so it is used as a backup
- B. The administrative distance must be higher on the primary route so that the backup route becomes secondary
- **C. The floating static route must have a higher administrative distance than the primary route so it is used as a backup**
- D. The default-information originate command must be configured for the route to be installed into the routing table

Question 4

Refer to the exhibit. Which route does R1 select for traffic that is destined to 192.168.16.2?

```
R1#show ip route
D    192.168.16.0/26 [90/2679326] via 192.168.1.1
R    192.168.16.0/24 [120/3] via 192.168.1.2
O    192.168.16.0/21 [110/2] via 192.168.1.3
i L1 192.168.16.0/27 [115/30] via 192.168.1.4
```

- **A. 192.168.16.0/27**
- B. 192.168.16.0/26
- C. 192.168.16.0/24
- D. 192.168.16.0/21

Explanation

The destination IP addresses match all four entries in the routing table but the 192.168.16.0/27 has the longest prefix so it will be chosen. This is called the “longest prefix match” rule.

Question 5

Which statement about static and dynamic routes is true?

- A. Static routes tell the router how to forward packets to networks that are not directly connected, while dynamic routes tell the router how to forward packets to networks that are directly connected
- B. Dynamic routes are manually configured by a network administrator, while static routes are automatically learned and adjusted by a routing protocol
- **C. Static routes are manually configured by a network administrator, while dynamic routes are automatically learned and adjusted by a routing protocol**
- D. Dynamic routes tell the router how to forward packets to networks that are not directly connected, while static routes tell the router how to forward packets to networks that are directly connected

Question 6

Which attribute does a router use to select the best path when two or more different routes to the same destination exist from two different routing protocols?

- A. dual algorithm
- B. hop count
- **C. administrative distance**
- D. metric

Question 7

Router A learns the same route from two different neighbors, one of the neighbor routers is an OSPF neighbor and the other is an EIGRP neighbor. What is the administrative distance of the route that will be installed in the routing table?

- A. 110
- B. 115
- **C. 90**
- D. 20

Explanation

The Administrative distance (AD) of EIGRP is 90 while the AD of OSPF is 110 so EIGRP route will be chosen to install into the routing table.

Question 8

Refer to the exhibit. With which metric was the route to host 172.16.0.202 learned?

```
R1#show ip route | begin gateway
Gateway of last resort is 209.165.200.246 to network 0.0.0.0
S* 0.0.0.0/0 [1/0] via 209.165.200.246, Serial0/1/0
     is directly connected, Serial0/1/0
     172.16.0.0/16 is variably subnetted, 3 subnets, 3 masks
S    172.16.0.0/24 [1/0] via 207.165.200.254, 00:00:33, Serial0/0/0
O    172.16.0.128/25 [110/38443] via 207.165.200.254, 00:00:33, Serial0/0/1
D    172.16.0.192/29 [90/3184439] via 207.165.200.254, 00:00:33, Serial0/0/1
     209.165.200.0/24 is variably subnetted, 4 subnets, 2 masks
C    209.165.200.248/30 is directly connected, Serial0/0/0
L    209.165.200.249/30 is directly connected, Serial0/0/0
C    209.165.200.252/30 is directly connected, Serial0/0/1
L    209.165.200.253/32 is directly connected, Serial0/0/1
```

- A. 0
- B. 110
- **C. 38443**
- D. 3184439

Explanation

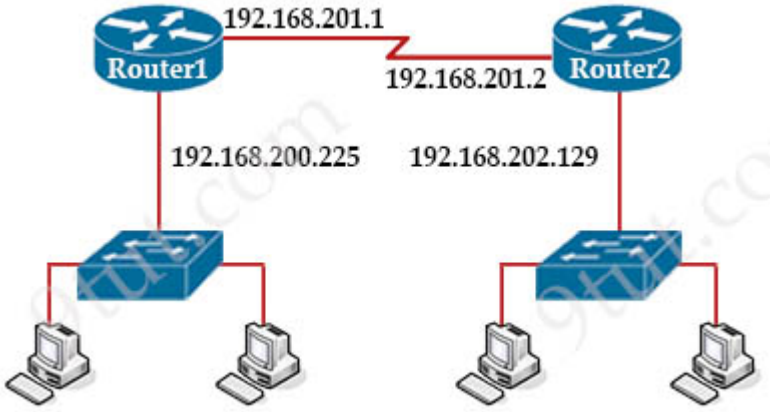
Both the line “O 172.16.0.128/25” and “S 172.16.0.0/24” cover the host 172.16.0.202 but with the “longest (prefix) match” rule the router will choose the first route.

Note: Network 172.16.0.192/29 does not cover host 172.16.0.202. For network 172.16.0.192/29:

Increment: 8
Network address: 172.16.0.192
Broadcast address: 172.16.0.199

Question 9

Refer to the exhibit. Which command would you use to configure a static route on Router 1 to network 192.168.202.0/24 with a nondefault administrative distance?



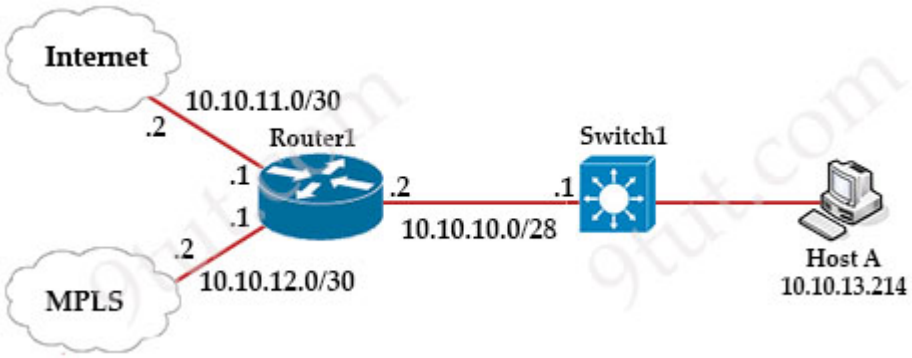
- A. router1(config)#ip route 1 192.168.201.1 255.255.255.0 192.168.201.2
- **B. router1(config)#ip route 192.168.202.0 255.255.255.0 192.168.201.2 5**
- C. router1(config)#ip route 5 192.168.202.0 255.255.255.0 192.168.201.2
- D. router1(config)#ip route 192.168.202.0 255.255.255.0 192.168.201.2 1

Explanation

The default AD of static route is 1 so we need to configure another number for the static route.

Question 10

Refer to the exhibit. Which prefix does Router 1 use for traffic to Host A?




```
Router1#show ip route
Gateway of last resort is 10.10.11.2 to network 0.0.0.0

   209.165.200.0/27 is sudnetted, 1 subnets
B     209.165.200.224 [20/0] via 10.10.12.2, 03:03:03
   209.165.201.0/27 is sudnetted, 1 subnets
B     209.165.201.0 [20/0] via 10.10.12.2, 03:03:03
   209.165.202.0/27 is subnetted, 1 subnets
B     209.165.202.128 [20/0] via 10.10.12.2, 03:03:03
   10.0.0.0/8 is variably subnetted, 8 subnets, 4 masks
C     10.10.10.0/28 is directly connected, GigabitEthernet0/0
C     10.10.11.0/30 is directly connected, FastEthernet2/0
C     10.10.12.0/30 is directly connected, GigabitEthernet0/1
O     10.10.13.0/25 [110/2] via 10.10.10.1, 00:00:03, GigabitEthernet0/0
O     10.10.13.128/28 [110/2] via 10.10.10.1, 00:00:03, GigabitEthernet0/0
O     10.10.13.144/28 [110/2] via 10.10.10.1, 00:00:03, GigabitEthernet0/0
O     10.10.13.160/29 [110/2] via 10.10.10.1, 00:00:03, GigabitEthernet0/0
O     10.10.13.208/29 [110/2] via 10.10.10.1, 00:00:03, GigabitEthernet0/0
S*  0.0.0.0/0 [1/0] via 10.10.11.2
```

- A. 10.10.10.0/28

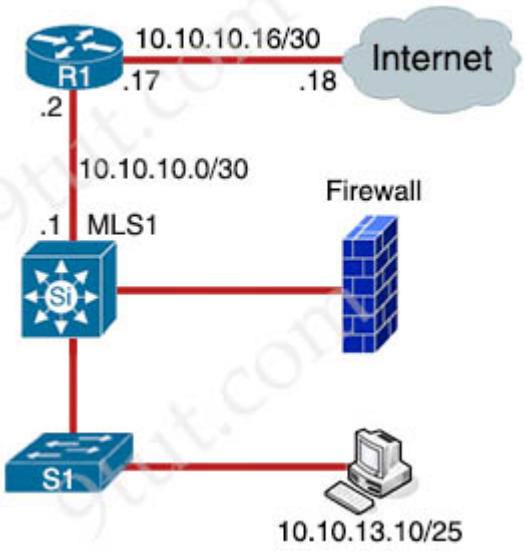
- **B. 10.10.13.208/29**
- C. 10.10.13.0/25
- D. 10.10.13.144/28

Explanation

The prefix with “longest prefix” will be matched first, in this case is “/29”.

Question 11

Refer to the exhibit. Which type of route does R1 use to reach host 10.10.13.10/32?




```
R1#show ip route
Gateway of last resort is 10.10.10.18 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 4 subnets. 3 Masks
C 10.10.10.0/30 is directly connected. FastEthernet0/1
O 10.10.13.0/25 [110/4576] via 10.10.10.1, 02:53:11, FastEthernet0/1
C 10.10.10.16/30 is directly connected, FastEthernet0/24
0 10.10.13.144/28 [110/110] via 10.10.10.1, 03:51:21, FastEthernet0/1
B* 0.0.0.0/0 [20/0] via 10.10.10.18, 02:17:53
```

- **A. network route**
- B. floating static route
- C. default route
- D. host route

Explanation

From the output, we see R1 will use the entry “O 10.10.13.0/25 [110/4576] via 10.10.10.1, …” to reach host 10.10.13.10. This is a network route.

Note: “B* 0.0.0.0/0 …” is a default route.

Question 12

Refer to the exhibit. How does router R1 handle traffic to 192.168.10.16?

```
R1# show ip route
D 192.168.10.0/24 [90/2679326] via 192.168.1.1
R 192.168.10.0/27 [120/3] via 192.168.1.2
O 192.168.10.0/28 [110/2] via 192.168.1.3
i L1 192.168.10.0/13 [115/30] via 192.168.1.4
```

- A. It selects the OSPF route because it has the lowest cost
- B. It selects the IS-IS route because it has the shortest prefix inclusive of the destination address
- **C. It selects the RIP route because it has the longest prefix: inclusive of the destination address**
- D. It selects the EIGRP route because it has the lowest administrative distance

Explanation

Note: The "O 192.168.10.0/28" route ranges from 192.168.10.0 to 192.168.10.15 only so it does not include destination 192.168.10.16 so OSPF route is not the correct answer.

Question 13

Refer to the exhibit. If configuring a static default route on the router with the **ip route 0.0.0.0 0.0.0.0 10.13.0.1 120** command, how does the router respond?

```
Gateway of last resort is 10.12.0.1 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 10.12.0.1, 00:00:01, GigabitEthernet0/0
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     10.0.0.0/24 is directly connected, GigabitEthernet0/0
L     10.0.0.2/32 is directly connected, GigabitEthernet0/0
C     10.13.0.0/24 is directly connected, GigabitEthernet0/1
L     10.13.0.2/32 is directly connected, GigabitEthernet0/1
```

- A. It starts sending traffic without a specific matching entry in the routing table to GigabitEthernet0/1
- B. It immediately replaces the existing OSPF route in the routing table with the newly configured static route
- **C. It ignores the new static route until the existing OSPF default route is removed**
- D. It starts load-balancing traffic between the two default routes

Explanation

Our new static default route has the Administrative Distance (AD) of 120, which is bigger than the AD of OSPF External route (O*E2) so it will not be pushed into the routing table until the current OSPF External route is removed.

For your information, if you don’t type the AD of 120 (using the command “ip route 0.0.0.0 0.0.0.0 10.13.0.1”) then the new static default route would replace the OSPF default route as the default AD of static route is 1. You will see such line in the routing table:

S* 0.0.0.0/0 [1/0] via 10.13.0.1

Question 14

Which of the following dynamic routing protocols are Distance Vector routing protocols? (Choose two)

- A. OSPF
- B. BGP

- **C. RIP**
- D. IS-IS
- **E. EIGRP**

Comments (8) Comments

1. Jalloh
   July 22nd, 2020

   11/12

   Slip up on 11 was user error. I overlooked the non-default AD of the static route.

2. Customer
   March 15th, 2021

   Refer to the exhibit. Which type of route does R1 use to reach host 10.10.13.10/32?

   Is the Prefix in the Question really correct?

3. anon
   May 27th, 2021

   @customer
   The correct answer is actually
   – O 192.168.10.0/28 [110/2] via 192.168.1.3
   RIP doesnt have the longest prefix

4. anon
   May 27th, 2021

   my mistake
   Explanation
   Note: The “O 192.168.10.0/28” route ranges from 192.168.10.0 to 192.168.10.15 only so it does not include destination 192.168.10.16 so OSPF route is not the correct answer.

5. I Take CCNA Tomorrow
   August 11th, 2022

   I call bs on the following:

   Refer to the exhibit. Which prefix does Router 1 use for traffic to Host A?

   I believe the correct answer is:

   10.10.13.0/25

6. Mau Ramirez
   November 15th, 2022

   So, Id like to confirm this as I do not recall reading about it in the actual book:

   When you have more than one matching route, I understand the first tie breaker is the admin distance. I see that here the longest prefix is used instead. Am I getting this right?

7. Anonymous
   December 13th, 2024

longest inclusive prefix always wins. Thats why “default routes”/0.0.0.0 are always of last resort, the router only sends it there if ther earent any specific, generally more local routes. For example if you set up vpn to your home to access some files there, you would add a specific route to your lan network. That way your device will check if the destination is in the small home lan and go across the vpn before sending it to the internet where it will get lost – but if the destination is not your home lan then it will send it will since most likely it will be a public ip excluded in your home but included in 0.0.0.0

8. Franziska
   September 23rd, 2025

   For this question:

   Refer to the exhibit. Which route does R1 select for traffic that is destined to 192.168.16.2?

   R1#show ip route
   D 192.168.16.0/26 [90/2679326] via 192.168.1.1
   R 192.168.16.0/24 [120/3] via 192.168.1.2
   O 192.168.16.0/21 [110/2] via 192.168.1.3
   i L1 192.168.16.0/27 [115/30] via 192.168.1.4

   you said the answer ist the /27 adress because of longest prefix BUT it is a level 1 address. wouldn’t the router pick a Level 2 address overa level 1 address? that would make the correct answer teh /26 address.

   any thoughts on that?

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Name

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- CCNA - New Questions Part 16
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- Voice VLAN Configuration Sim 2
- Voice VLAN & LLDP Sim
- Allowed, Native VLAN & LACP Sim
- Static Routing Configuration Sim 4
- IPv4 and IPv6 Connectivity Sim
- VLAN & CDP Sim 2
- IPv4 and IPv6 Assignment Sim
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- OSPF Configuration Sim 2
- OSPF Configuration Sim 3
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- VLAN Configuration Sim
- Allowed, Native VLAN & LACP Sim 2
- 802.1Q Trunking & LACP
- IPv4 and IPv6 Assignment Sim 2
- 802.1Q Trunking, Native VLAN & LACP
- Static Routing Configuration Sim 5
- Static Routing Configuration Sim 6
- PAT, NTP & DHCP Relay Sim
- OSPF DR Sim
- NACL DHCP Snooping 3
- NAT DHCP SSH Sim
- ==== Lab Sims for CCNA (although they are rather old but please learn all of them too) ====
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- LACP Configuration Sim
- Voice VLAN Configuration Sim
- VLAN and Trunking Configuration Sim
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- Named Access-list & DHCP Snooping Sim
- VLAN and CDP Sim
- IPv4 and IPv6 Deployment Sim
- Static Routing Configuration Sim 3
- VLAN & LLDP Sim
- IPv6 Static Routing Sim
- Named Access-list & DHCP Snooping Sim 2
- ISL Trunking and LACP Sim
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- Static Routing Configuration Sim 2
- IP Services Sim
- Static Routing Configuration Sim

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- OSPF Questions
- OSPF Questions 2
- EIGRP Questions
- NAT Questions
- NTP Questions

- Syslog Questions
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